

Individual Contribution Reflection — Raven Ge (wege8390)

COMP4347 / COMP5347 Assignment 2 · Quiz subsystem, Review Mode, frontend integration and UI/UX ownership

1. Contribution Summary

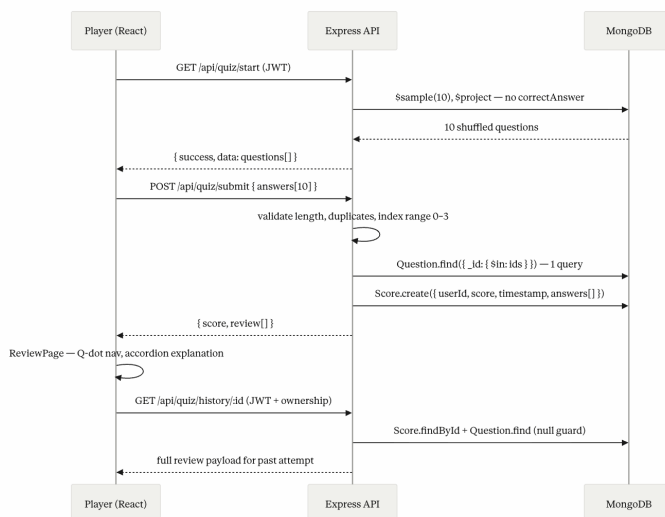
I owned the quiz subsystem across backend and frontend. On the backend (def48eb), I implemented `quiz.controller.js` in full: randomised question selection via MongoDB `$sample` fixed at 10, input validation (answer count, duplicate detection, index range 0–3), batch-fetch scoring replacing an N+1 loop, server-side `review[]` assembly for Review Mode, leaderboard `$lookup` join, and history retrieval with ownership validation. On the frontend (62ca283), I built `QuizContext.js` (useReducer state management), `Quiz.js`, `AttemptReview.js` (Review Mode page), and `History.js`. These files were lost in a subsequent merge conflict and are verifiable via `git show 62ca283`. During the integration phase I took primary ownership of the player-facing UI/UX system, including gameplay screens, Review Mode layouts, navigation components, and the Gothic Academia visual theme.

2. Major Technical Challenge

The original `submitQuiz` called `Question.findById()` inside a loop, producing ten sequential database queries per submission. I refactored to a single `Question.find({ _id: { $in: questionIds } })` call with a hashmap lookup, reducing database round trips from 10 to 1. The same hashmap then powered Review Mode assembly at no extra cost, returning `questionText`, `options`, `correctAnswer`, and `explanation` in the submit response without additional queries.

A second challenge emerged after adding `shuffleQuestion.js` (41f7ce8): the submitted `selectedAnswer` index must correspond to the shuffled option order, not the seed order. I resolved this by preserving the shuffled array throughout the quiz lifecycle so that scoring and review reconstruction remained consistent with what the user actually saw during gameplay.

3. Subsystem Sequence Diagram (Mermaid)



4. Git Commit Analysis

Hash	What
def48eb	Rewrote <code>quiz.controller.js</code> : batch-fetch scoring ($N+1 \rightarrow 1$ query), input validation, <code>review[]</code> assembly, leaderboard \$lookup, history ownership check (+189 lines)
62ca283	Quiz frontend: <code>QuizContext.js</code> (useReducer), <code>Quiz.js</code> , <code>AttemptReview.js</code> (Review Mode), <code>History.js</code> . <i>Files lost in merge conflict — verifiable via <code>git show 62ca283</code></i>
8724146	Major UI overhaul: <code>QuizScreens.jsx</code> (+601), <code>quizflow.css</code> (+1654), Sydney SVG assets, <code>QuizWorldBackground.jsx</code> (16 files, +2997 lines)
e22b914	Background assets added, <code>QuizScreens.jsx</code> interaction fixes, <code>quizflow.css</code> refactor (4 files)
4c1730d	Quiz flow animation and interaction refinements (2 files, +156 –134)
268d0c6	Review Mode layout, History page styling, <code>quizflow.css</code> +1698 (5 files, +1952 lines)
41f7ce8	Added <code>shuffleQuestion.js</code> , updated <code>quiz.controller.js</code> and <code>Question.js</code> model, seed data (+655 lines), <code>ReviewPage.jsx</code> +192 (16 files, +1873 lines)
d17fe3c	Quiz flow logic fixes: <code>QuizScreens.jsx</code> +380, <code>HistoryPage.jsx</code> +101, login/register UI, brand assets (26 files, +1756 lines)
657d15b	Added <code>forbidAdminQuiz.middleware.js</code> + unit test, <code>Leaderboard.jsx</code> +87, <code>HistoryPage.jsx</code> +158, <code>ReviewPage.jsx</code> +49 (25 files)
0102a29	<code>ReviewV7Card.jsx</code> , <code>reviewFormatUtils.js</code> , <code>HistoryPage.jsx</code> +282, <code>ReviewPage.jsx</code> +222, <code>quiz.controller.js</code> +48, new CSS modules (20 files, +3572 lines)
d04b018	Navigation system: <code>PlayerNavbar.jsx</code> , <code>AdminNavbar.jsx</code> , <code>GameplayHudPortal.jsx</code> , <code>QuizScreens.jsx</code> +691, typography + gothic CSS (41 files, +5325 lines)
93a9096	CSS quality pass: <code>quizflow.css</code> +107, <code>quiz-play.css</code> +47 (7 files)
64d01cf	Landing page fix, <code>ProtectedAdminRoute.jsx</code> , <code>motion-system.css</code> +298 (23 files)
c5ea913	Leaderboard tie-breaking: users with equal scores ranked by earliest submission timestamp; result set capped at 50 entries

5. Review Mode Design Reflection

The `review[]` array is assembled server-side during submission using the `questionMap` already loaded for scoring, avoiding an extra API call and keeping the Score model minimal. I replaced an initial flip-card design with accordion-based explanations because flip-cards imply hidden information — appropriate during gameplay, not after submission when all answers are already revealed. The empty-string guard (`if (q.explanation)`) suppresses the expand button when no explanation exists, since MongoDB stored the field as `""` rather than `absent`. Finally, because `shuffleQuestion.js` randomises option order per attempt, the review payload must preserve the same shuffled ordering shown during gameplay — failing to do so would produce a review screen that appears correct but explains the wrong option.